Cureus
Part of SPRINGER NATURE

Text Sentiment Analyser: A Hybrid Rule-Based and Machine Learning Framework for Sentiment and Emotion Analysis of English and Code-Mixed Hinglish Text

Sudhanshu Shrivastav, Ravi Kumar Bhardwaj and Audarsh Singh Rathore

Department of Computer Science and Engineering, Maharana Institute of Professional Studies, Kanpur, Uttar Pradesh 209217, India (affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow)

Abstract. Indian social-media users often type Hindi in the Latin alphabet and switch to English within one sentence, a style known as Hinglish. English sentiment tools handle such text poorly because Romanised Hindi opinion words are missing from their lexicons, spellings vary widely and Hindi places the negator after the word it negates. This paper presents Text Sentiment Analyser, a privacy-preserving system that labels English and Hinglish text as Positive, Neutral or Negative and profiles its emotions. The design joins four parts: a language identifier with a spelling normaliser; a readable bilingual word list whose rules cover negation placed before or after a word, words such as *bahut* that strengthen an opinion, and contrast words such as *lekin*; two trained classifiers (multinomial Naive Bayes and logistic regression on TF-IDF word pairs); and a weighted blend of the rule scores with Naive Bayes. Extra modules report Plutchik's eight emotions, opinions about named product aspects, possible sarcasm and the words that drove each decision. We tested five engines on 210 balanced, hand-labelled texts using five-fold stratified cross-validation. Accuracy was 55.2% for an English-only word list, 92.4% for our Hinglish-aware rules (95% CI 88.6–95.7), 65.9% for Naive Bayes and 82.1% for the blend, averaged over five random splits; the blend beat Naive Bayes significantly (McNemar $p < 0.001$). Since our word list and test texts were prepared together, the 92.4% figure is best treated as a ceiling. Ablation, learning-curve and error studies show which parts matter. The tool works offline inside a browser.

Keywords: *Hinglish, Code-mixing, Opinion mining, Rule-based classifier, Naive Bayes, Emotion recognition, Explainable NLP, Text classification*

Introduction:

A. Background and Motivation

Sentiment analysis is the computational study of the opinions and feelings people express in text, and it supports review mining, brand monitoring, complaint triage and opinion research [1–3]. Its best-known tools, including VADER [4], SO-CAL [5] and the supervised classifiers of Pang et al. [6], were built for English. Many Indian users, however, switch between Hindi and English within a sentence and type Hindi with a Roman keyboard [7], which gives comments like *"movie acchi nahi thi but songs mast the"*, meaning that the film disappointed but its songs were good.

Such text defeats English tools in three ways. Hindi opinion words such as *bakwas* (rubbish) and *zabardast* (excellent) are absent from English lexicons; Romanised Hindi has no standard spelling, so *bahut* (very) also appears as *bahot*, *bhot* or *bohot*; and Hindi places the negator after the word it negates (*acchi nahi thi*), so English negation windows point the wrong way. Code-mixing is known to harm language identification, tagging and sentiment classification [8–10].

B. Problem Statement and Objectives

Tools built for English mark a large share of Hinglish comments as Neutral, and trained models would need labelled Hinglish examples, which a student project seldom has in quantity. We therefore set out (i) to build a rule engine that understands Hinglish spellings, negation in both languages, intensifying words and contrast words; (ii) to train Naive Bayes and logistic regression on labelled examples; (iii) to blend the rules with Naive Bayes and find out when blending pays off; (iv) to report emotions, aspects, sarcasm hints and word-level reasons; and (v) to evaluate all engines with reproducible statistics. Our contributions are a pipeline with no outside libraries whose resources hold 292 English and 184 Hinglish opinion words, 28 Hindi intensifiers and 94 spelling-variant pairs, a five-engine comparison with confidence intervals and significance tests, and ablation, learning-curve and error analyses.

Literature Review:

[3] Birjali et al. (2021) and [11] Wankhade et al. (2022) surveyed lexicon-based, machine-learning and hybrid sentiment methods and identified negation, sarcasm and code-mixed, low-resource text as open problems. [12] Nandwani and Verma (2021) reviewed sentiment analysis together with emotion detection, and [13] Bordoloi and Biswas (2023) proposed a design framework, from pre-processing to visualisation, that matches the pipeline used here.

[4] Hutto and Gilbert (2014) proposed VADER, a validated lexicon with heuristics for punctuation, capitals, degree words, *but* and negation, normalised into a compound score that our engine adopts; [5] Taboada et al. (2011) modelled negation as an attenuating shift of polarity. [14] Catelli et al. (2022) found BERT more accurate than a lexicon on Italian but the lexicon faster, data-free and interpretable. [6] Pang et al. (2002) and [15] Wang and Manning (2012) showed that Naive Bayes and linear models over unigrams and bigrams are strong baselines, and [16] Tan et al. (2022) combined RoBERTa with an LSTM for higher accuracy on large English corpora.

[17] Doğruöz et al. (2021) surveyed code-switching from linguistic and social perspectives. [18] Khanuja et al. (2020) built the GLUECoS benchmark for code-switched NLP, and [19] Nayak and Joshi (2022) pre-trained HingBERT on a large Hinglish corpus. [20] Joshi et al. (2016) introduced a sub-word LSTM for Hindi-English comments that was 4–5% more accurate than traditional methods, [10] Patra et al. (2018) fixed the three-class task in the SAIL shared task, and [21] Patwa et al. (2020) organised SemEval-2020 SentiMix, where top systems such as that of [22] Srinivasan (2020) fine-tuned multilingual transformers [23]. [24] Singh and Lefever (2020) transferred English knowledge through cross-lingual embeddings, whereas [25] Baroi et al. (2020) obtained an F-score of only 0.617 with a classical ensemble.

[26] Chakravarthi et al. (2022) released DravidianCodeMix and highlighted class imbalance and annotation cost. [27] Mehmood et al. (2020) showed that normalising Roman Hindi and Urdu spellings improves sentiment classification, and [28] Mabokela et al. (2023) concluded that lexicons and normalisation remain vital for under-resourced languages. [29] Plutchik (1980) proposed eight basic emotions, used in the NRC lexicon of [30] Mohammad and Turney (2013). [31] Ribeiro et al. (2016) introduced perturbation-based explanation and [32] Ribeiro et al. (2020) exposed negation failures in commercial sentiment models through behavioural tests. Finally, [33] Zhang et al. (2024) and [34] Zhang et al. (2023) found that large language models lag on fine-

grained and code-switched tasks and require network access. Table 1 summarises the most relevant studies.

Table 1. Literature review summary chart of the studies most relevant to this work (SN = serial number; the number in square brackets is the reference number).

SN	Name of Authors & Reference Number	Title of Paper	Objectives	Methodology	Outcome	Journal / Conference
1	Hutto & Gilbert (2014) [4]	VADER: A parsimonious rule-based model for sentiment analysis of social media text	Build a fast, general rule-based analyser for social-media text	Validated lexicon with five heuristics (punctuation, capitals, degree words, “but”, negation); compound score	Outperformed individual human raters on tweets; widely adopted baseline; English only	ICWSM (AAAI), 2014
2	Joshi et al. (2016) [20]	Towards sub-word level compositions for sentiment analysis of Hindi-English code mixed text	Classify sentiment of Hindi-English Facebook comments	New dataset; sub-word LSTM	4–5% above traditional methods and 18% above prior Hi-En system; needs labelled data	COLING, 2016
3	Patra et al. (2018) [10]	Sentiment analysis of code-mixed Indian languages: Overview of SAIL_Code-Mixed shared task	Benchmark three-class sentiment for Hi-En and Bn-En tweets	Shared task with released data and common evaluation	Nine teams participated; defines the three-class task used here	arXiv / ICON, 2018
4	Patwa et al. (2020) [21]	SemEval-2020 Task 9: Overview of sentiment analysis of code-mixed tweets	Create a benchmark for Hinglish and Spanglish sentiment	Large annotated tweet corpora; shared task	Top systems fine-tuned multilingual transformers; standard benchmark	SemEval (ICCL), 2020
5	Mehmood et al. (2020) [27]	An unsupervised lexical normalization for Roman Hindi and Urdu sentiment analysis	Normalise spelling variants of Roman Hindi/Urdu	Unsupervised clustering of lexical variants	Normalisation improved sentiment classification	Information Processing & Management, 2020
6	Nayak & Joshi (2022) [19]	L3Cube-HingCorpus and HingBERT	Provide Hinglish pre-training data and models	Large unlabelled corpus; BERT pre-training	HingBERT beat general multilingual BERT on code-mixed tasks	WILDRE-6 @ LREC, 2022
7	Ribeiro et al. (2020) [32]	Beyond accuracy: Behavioral testing of NLP models with CheckList	Test model behaviour beyond accuracy	Capability-based test suites (e.g., negation)	Found critical failures in commercial sentiment models	ACL, 2020
8	Mabokela et al. (2023) [28]	Multilingual sentiment analysis for under-resourced languages: A systematic review	Map research on low-resource sentiment analysis	Systematic literature review	Lexicons and normalisation remain vital where data are scarce	IEEE Access, 2023
9	Zhang et al. (2023) [34]	Multilingual large language models are not (yet) code-switchers	Evaluate LLMs on code-switching tasks	Benchmarking multilingual LLMs vs fine-tuned models	LLMs underperform smaller fine-tuned models on code-switched data	EMNLP, 2023

Research gap. Rule-based tools are English-only, strong code-mixed systems need large labelled corpora and GPUs, few systems explain verdicts word by word, and small studies seldom report confidence intervals or ablations. The present work addresses these gaps.

Methods and Materials:

A. Architecture and Corpus

Text enters by keyboard, voice or CSV upload (Fig. 1). A language identifier tags it as English, Roman Hindi, Hinglish or Devanagari, and a normaliser collapses repeated letters and maps spelling variants to canonical forms. The cleaned tokens go to the rule engine and to both classifiers at the same time, after which a blending step returns a verdict with a confidence value; separate modules compute emotions, aspect-level sentiment, a sarcasm indicator, a word-level explanation and text statistics. The system has about 4,300 lines of JavaScript and runs in a browser or Node.js; a Python script reproduces the experiments with scikit-learn [35].

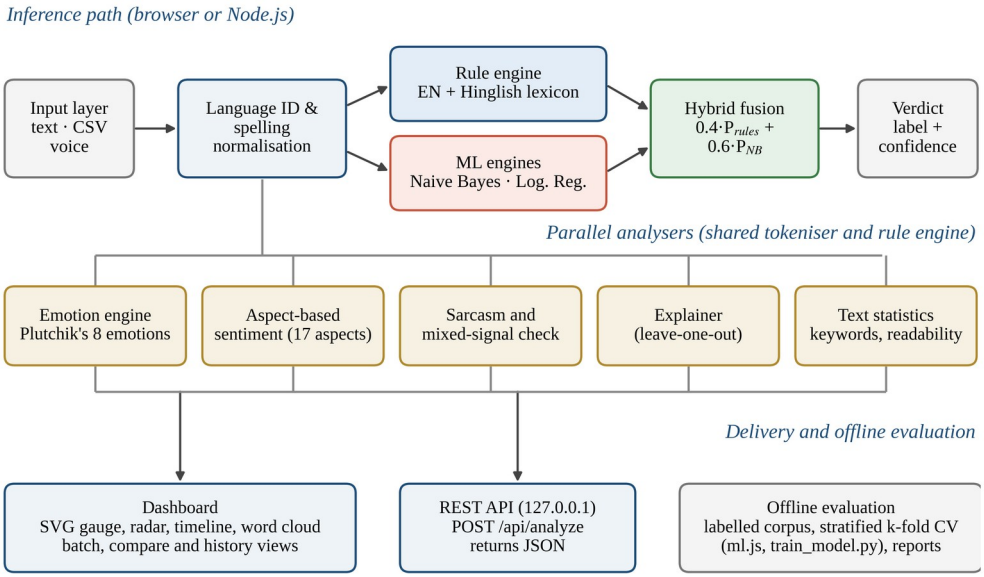


Fig. 1. Architecture of Text Sentiment Analyser: inference path (top), parallel analysers (middle), and delivery and offline evaluation (bottom).

Our test set contains 210 short English and Hinglish sentences that we wrote ourselves, modelled on public comments about films, phones, cricket, food, travel, college and customer service (Positive 71, Neutral 69, Negative 70; mean length 7.1 words; 133 Hinglish, 63 Roman Hindi and 14 English). When a comment mixes opinions we labelled it by its main or closing opinion, sarcastic comments by what the writer really meant, and plain facts or questions as Neutral. Because the Hinglish word list was compiled alongside the corpus, the rule engine's score is an optimistic upper bound; the learned models are always tested on held-out folds.

B. Normalisation and Rule-Based Engine

The language identifier counts tokens from a 211-word Romanised-Hindi list and an English function-word list (Hinglish when the Hindi share reaches 15%). The normaliser lower-cases

tokens, reduces runs of three or more identical letters to two (*bahuuut → bahuut*) and applies 94 variant mappings (*bhot*, *bohot → bahut*; *nhi → nahi*).

The rule engine scans tokens once, accumulating positive evidence P and negative evidence N (Table 2). Lexicon words carry valences of 1–4 and emoji −4 to +4. An English negator reverses and attenuates sentiment words in the next three tokens by −0.75 [5]; a Hindi negator instead looks back up to three words and multiplies the most recent sentiment score by −0.75, so *acchi nahi thi* becomes negative, while emoji are never reversed. Boosters scale the next word (*bahut* ×1.6, *thoda* ×0.6), and a contrast word (*but*, *lekin*, *magar*) halves earlier evidence and multiplies later evidence by 1.5. Eq. (1) gives a token's score and Eq. (2) the compound score [4]; the text is called Positive when c is at least 0.05 with P above N, Negative when c is at most −0.05 with N above P, and Neutral in every other case.

$$s_t = v_t \cdot b_t \cdot \eta_t \cdot \kappa_t \quad (1)$$

$$c = x / \sqrt{(x^2 + 15)}, \quad x = P - N \quad (2)$$

In Eq. (1), v_t is the word's stored strength and b_t its booster multiplier; η_t is −0.75 when the word is negated and 1 if not, and κ_t is 1.5 for words after a contrast word and 1 if not.

Table 2. Main parameters of the rule engine and linguistic resources.

Parameter / resource	**Value**
English / Hinglish lexicon	292 / 184 entries, valence 1–4; emoji −4 to +4
Negation	Factor −0.75; English forward window and Hindi look-back of 3 tokens
Intensifiers	28 Hindi and 27 English; factors 0.5–1.8
Contrast words (but, lekin, magar)	×0.5 before, ×1.5 after
Emphasis	ALL-CAPS ×1.3; each "!" +8% (at most four)
Emotion lexicon and aspects	376 word–emotion entries (8 emotions); 17 aspect categories

C. Machine Learning and Hybrid Fusion

Both classifiers use normalised unigrams and bigrams, keeping emoji, "!" and "?" as tokens. Multinomial Naive Bayes [36] uses Laplace smoothing (Eq. 3, $\alpha = 1$). Logistic regression uses sub-linear TF-IDF vectors [37] with a softmax output trained by stochastic gradient descent with L2 regularisation (40 epochs, $\eta_0 = 0.4$, $\lambda = 10^{-4}$). The hybrid converts c into probabilities and mixes them with the Naive Bayes posterior (Eqs. 4–5); the label is the arg-max.

$$P(f \mid y) = (n_{f,y} + \alpha) / (N_y + \alpha|V|) \quad (3)$$

$$P_r(\text{Pos}) = \max(c, 0), \; P_r(\text{Neg}) = \max(-c, 0), \; P_r(\text{Neu}) = 0.9 \cdot \max(0, 1 - |c|) + 0.05 \quad (4)$$

$$P_{\text{hybrid}}(y) = w \cdot P_r(y) + (1 - w) \cdot P_{\text{NB}}(y), \quad w = 0.4 \quad (5)$$

The emotion module assigns lexicon evidence to Plutchik's eight emotions [29]; the aspect module scores clauses containing keywords such as *camera* or *battery*; the sarcasm module reports rule-based cues (eye-roll emoji, *wah kya baat hai*); and the explainer measures each word's impact by deleting it and rescoring, $\Delta_i = c(T) - c(T \setminus t_i)$ [31]. All charts are hand-written SVG, a content-security policy blocks outgoing requests, and the optional REST API listens on 127.0.0.1 only.

D. Evaluation Protocol

Engines are evaluated by stratified five-fold cross-validation [38], with the primary split at seed 7 and repeats at seeds 11, 21, 42 and 99. Metrics are accuracy, macro-F1, Cohen's kappa [39] and

95% confidence intervals from 2,000 bootstrap resamples [40]; engines are compared with McNemar's test [41], and kappa is read on the Landis–Koch scale [42]. Ablations remove one rule component at a time, and further runs sweep w and vary the training size. No setting was tuned to improve headline figures.

Results:

A. Overall Performance

The headline figures are in Table 3 and Fig. 2. With English words alone the engine scores 55.2%, mostly because it calls 76% of all texts Neutral. The Hinglish layer raises accuracy to 92.4% (κ = 0.89), a 37.2-point gain. Naive Bayes and logistic regression average 65.9% over five splits, and the hybrid 82.1%, 16.2 points above Naive Bayes. The scikit-learn models behave similarly (Naive Bayes 62.4%, logistic regression 65.7%, linear SVM 64.8%), which suggests that, with roughly 170 sentences to learn from, the shortage of data matters more than which classifier is used.

Table 3. Performance of the five engines (stratified five-fold cross-validation, n = 210; primary split except the last column).

Engine	Accuracy (%)	Macro-F1 (%)	95% CI (%)	κ	Five-split mean (%)
English lexicon (baseline)	55.3 ± 6.4	52.8 ± 8.0	48.6–61.9	0.33	55.2 ± 0.0
Hinglish-aware rules (ours)	92.4 ± 2.4	92.4 ± 2.2	88.6–95.7	0.89	92.4 ± 0.0
Naive Bayes	70.0 ± 3.1	69.9 ± 3.6	63.8–75.7	0.55	65.9 ± 2.8
Logistic regression	67.6 ± 2.8	67.3 ± 2.9	61.4–73.3	0.51	65.9 ± 1.8
Hybrid: rules + NB (ours)	83.8 ± 4.7	83.7 ± 4.6	78.6–88.6	0.76	82.1 ± 1.6

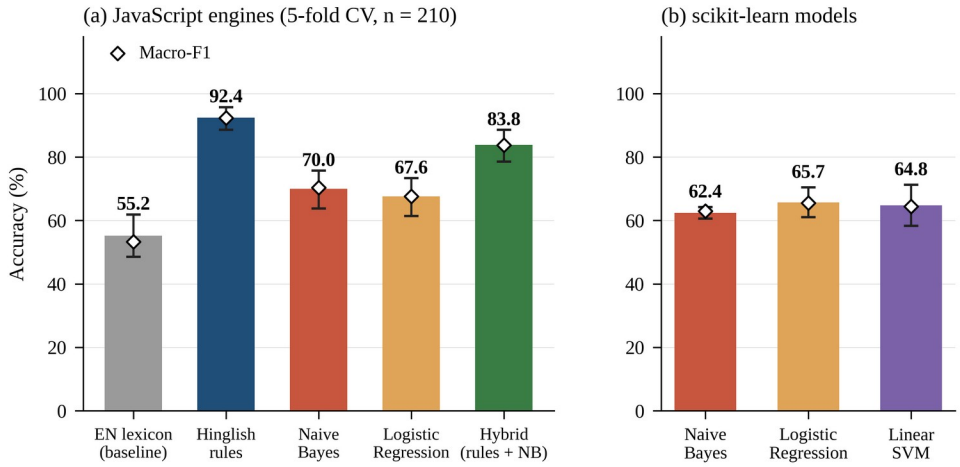


Fig. 2. Accuracy of the compared engines: (a) JavaScript engines with 95% bootstrap intervals; (b) scikit-learn models (mean ± SD over five folds). Diamonds show macro-F1.

B. Statistical Reliability and Class Behaviour

Bootstrap intervals are wide (half-widths 3.6–6.7 points), so only large differences are reliable. Paired McNemar tests show the Hinglish rules ahead of the English baseline (81 texts against 3, $p < 0.0001$), the blend ahead of Naive Bayes (30 against 1) and of logistic regression (35 against 1, both $p < 0.0001$), and the rules ahead of the blend (24 against 6, $p = 0.002$), whereas Naive Bayes and

logistic regression cannot be separated (p = 0.27). All engines except the baseline recognise Neutral texts well (F1 0.87–0.92), but the learned models confuse Positive and Negative (Fig. 3) because unseen opinion words contribute nothing at test time, while the Hinglish engine mainly misses complaints that contain no lexicon word.

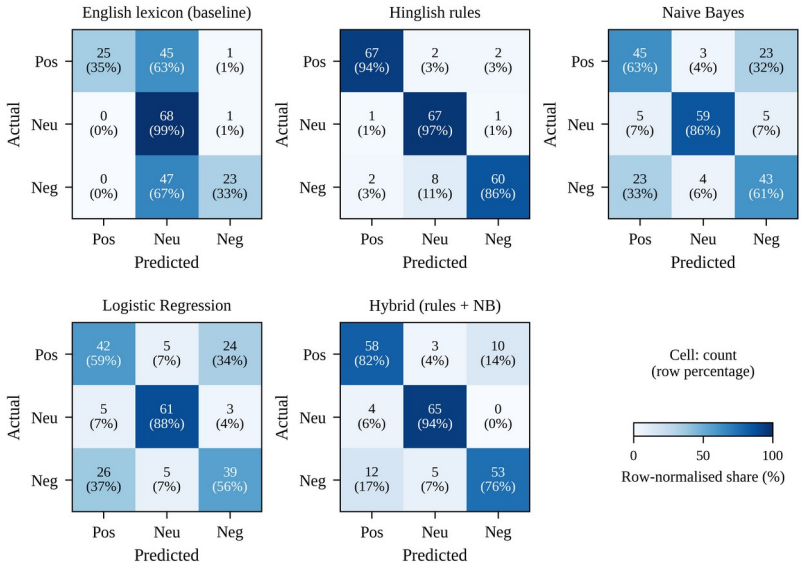


Fig. 3. Confusion matrices of the five engines (pooled out-of-fold predictions; rows are actual classes).

C. Ablation, Sensitivity and Data Efficiency

Removing the Hindi lexicon drops accuracy from 92.4% to 55.2% (Table 4). Taking out any other single rule moves overall accuracy by one point or less, but it does break hand-picked test sentences like *movie acchi nahi thi* and *camera mast hai but battery bekaar*, because the corpus contains few negators and contrast words [32]. Post-negation corrects four texts and breaks four, two of them "never … such" idioms with *kabhi nahi*. Varying w (Fig. 4a) gives a maximum of 89.4% at w = 0.7, an optimistic bound since it is chosen on the test folds. With only 17 training texts, Naive Bayes scores 49.9% but the hybrid already 75.5% (Fig. 4b), showing that the rules supply prior knowledge when data are scarce.

Table 4. Ablation of the Hinglish rule engine (Δ in percentage points; last column: 12 targeted test sentences).

Variant	Accuracy (%)	Δ	Texts changed	Tests passed
Full Hinglish engine	92.4	—	0	12/12
− Hindi post-negation	92.4	0.0	8	9/12
− Contrast rule (but, lekin)	92.4	0.0	0	10/12
− Intensifiers (bahut, ekdum)	91.9	−0.5	1	12/12
− English pre-negation	92.4	0.0	0	10/12
− Emoji and emoticon scores	91.4	−1.0	2	11/12
− Hindi lexicon (English only)	55.2	−37.1	86	5/12

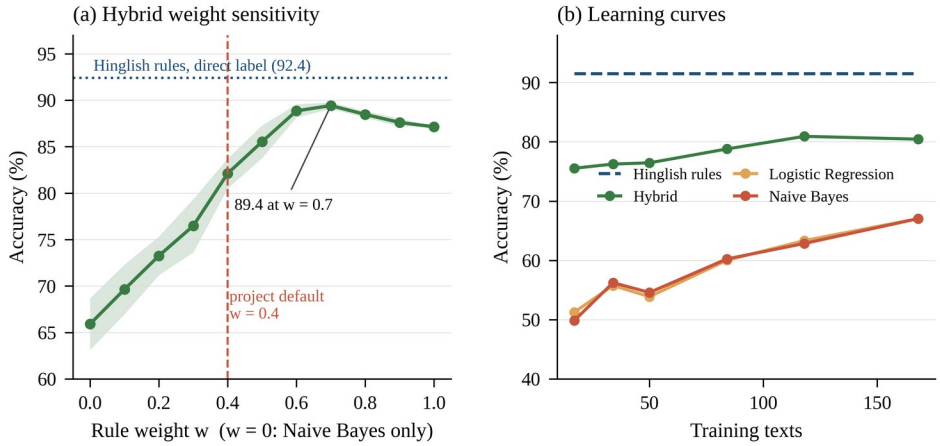


Fig. 4. (a) Hybrid accuracy as the rule weight w varies; (b) learning curves of the learned models, the hybrid and the rule engine.

D. Error Analysis and Runtime

The 16 errors of the Hinglish engine fall into three groups: nine implicit complaints with no lexicon word (*Battery bahut jaldi khatam ho jaati hai*), four negation-scope errors involving *kabhi nahi* or comparatives, and three domain-dependent words (*problem*, *jeet*, *refund*). The learned models fail mainly on unseen vocabulary; Naive Bayes labels *Heroine ki acting lajawab thi* Negative because *lajawab* never occurs in training. On *movie acchi nahi thi but songs mast the* the engine gives +0.70 (Positive), and the explanation view shows *mast* (+0.89) and *but* (+0.34) doing most of the work. All automated tests pass, and one text is analysed in 0.12 ms on average.

Figures and Tables

Note: all figures were produced at 600 dpi with labelled axes, and the first row of every table is a header row.

Discussion:

The central finding is that, when labelled data are scarce, vocabulary coverage rather than algorithm choice decides accuracy on Hinglish: lexicon entries added 37 points, while moving between Naive Bayes, logistic regression and a linear SVM shifted results by just a few points, consistent with earlier observations on linear baselines [15] and under-resourced languages [28]. The hybrid improved clearly on Naive Bayes, especially with little data, but did not beat the rule engine, partly because the lexicon was built with the corpus and partly because Eq. (4) suppresses weak rule signals while Naive Bayes posteriors are extreme. Corpus accuracy also hid the value of rules that fixed targeted cases, so challenge sets should accompany any benchmark [32].

The system is transparent, fast (about 0.1 ms per text) and fully offline, which suits privacy-sensitive uses and makes it a baseline for heavier models, including large language models [33, 34]. Its limitations are a small, team-written and balanced corpus, an optimistic rule-based score, unmeasured inter-annotator agreement, a fixed hybrid weight, unscored emotion and sarcasm modules, and Roman-script-only input. Future work will collect at least 1,000 real doubly annotated comments, tune w by nested validation, and compare against MuRIL-style Indic transformers and HingBERT [19].

Conclusion:

We built Text Sentiment Analyser to read opinions and emotions in English and Hinglish without sending any text off the user's machine, using hand-written language rules alongside trained classifiers. On our 210 test sentences, the Hinglish-aware engine reached 92.4% accuracy against 55.2% for an English-only lexicon, and the hybrid reached 82.1%, which is about 16 points better than Naive Bayes, and the gap widens when there is less training data.

We wrote the word list and the test sentences together, so the rule score should be read as a best case, and 210 sentences cannot stand in for real social-media traffic. What the work does offer is a design whose every decision can be traced, an evaluation that others can repeat, and evidence about which parts actually help. Our next step is to test it on real comments labelled by two independent annotators.

References:

[1] Liu, B. (2012). Sentiment analysis and opinion mining. *Synthesis Lectures on Human Language Technologies, 5*(1), 1–167. https://doi.org/10.2200/S00416ED1V01Y201204HLT016

[2] Pang, B., & Lee, L. (2008). Opinion mining and sentiment analysis. *Foundations and Trends in Information Retrieval, 2*(1–2), 1–135. https://doi.org/10.1561/1500000011

[3] Birjali, M., Kasri, M., & Beni-Hssane, A. (2021). A comprehensive survey on sentiment analysis: Approaches, challenges and trends. *Knowledge-Based Systems, 226*, 107134. https://doi.org/10.1016/j.knosys.2021.107134

[4] Hutto, C. J., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media, 8*(1), 216–225. https://doi.org/10.1609/icwsm.v8i1.14550

[5] Taboada, M., Brooke, J., Tofiloski, M., Voll, K., & Stede, M. (2011). Lexicon-based methods for sentiment analysis. *Computational Linguistics, 37*(2), 267–307. https://doi.org/10.1162/COLI_a_00049

[6] Pang, B., Lee, L., & Vaithyanathan, S. (2002). Thumbs up? Sentiment classification using machine learning techniques. In *Proceedings of the 2002 Conference on Empirical Methods in Natural Language Processing* (pp. 79–86). Association for Computational Linguistics. https://doi.org/10.3115/1118693.1118704

[7] Bali, K., Sharma, J., Choudhury, M., & Vyas, Y. (2014). "I am borrowing ya mixing?" An analysis of English-Hindi code mixing in Facebook. In *Proceedings of the First Workshop on Computational Approaches to Code Switching* (pp. 116–126). Association for Computational Linguistics. https://doi.org/10.3115/v1/W14-3914

[8] Barman, U., Das, A., Wagner, J., & Foster, J. (2014). Code mixing: A challenge for language identification in the language of social media. In *Proceedings of the First Workshop on Computational Approaches to Code Switching* (pp. 13–23). Association for Computational Linguistics. https://doi.org/10.3115/v1/W14-3902

[9] Vyas, Y., Gella, S., Sharma, J., Bali, K., & Choudhury, M. (2014). POS tagging of English-Hindi code-mixed social media content. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing* (pp. 974–979). Association for Computational Linguistics. https://doi.org/10.3115/v1/D14-1105

[10] Patra, B. G., Das, D., & Das, A. (2018). *Sentiment analysis of code-mixed Indian languages: An overview of SAIL_Code-Mixed shared task @ICON-2017* (arXiv:1803.06745). arXiv. https://doi.org/10.48550/arXiv.1803.06745

[11] Wankhade, M., Rao, A. C. S., & Kulkarni, C. (2022). A survey on sentiment analysis methods, applications, and challenges. *Artificial Intelligence Review, 55*(7), 5731–5780. https://doi.org/10.1007/s10462-022-10144-1

[12] Nandwani, P., & Verma, R. (2021). A review on sentiment analysis and emotion detection from text. *Social Network Analysis and Mining, 11*(1), 81. https://doi.org/10.1007/s13278-021-00776-6

[13] Bordoloi, M., & Biswas, S. K. (2023). Sentiment analysis: A survey on design framework, applications and future scopes. *Artificial Intelligence Review, 56*(11), 12505–12560. https://doi.org/10.1007/s10462-023-10442-2

[14] Catelli, R., Pelosi, S., & Esposito, M. (2022). Lexicon-based vs. BERT-based sentiment analysis: A comparative study in Italian. *Electronics, 11*(3), 374. https://doi.org/10.3390/electronics11030374

[15] Wang, S., & Manning, C. D. (2012). Baselines and bigrams: Simple, good sentiment and topic classification. In *Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)* (pp. 90–94). Association for Computational Linguistics.

[16] Tan, K. L., Lee, C. P., Anbananthen, K. S. M., & Lim, K. M. (2022). RoBERTa-LSTM: A hybrid model for sentiment analysis with transformer and recurrent neural network. *IEEE Access, 10*, 21517–21525. https://doi.org/10.1109/ACCESS.2022.3152828

[17] Doğruöz, A. S., Sitaram, S., Bullock, B. E., & Toribio, A. J. (2021). A survey of code-switching: Linguistic and social perspectives for language technologies. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing* (pp. 1654–1666). https://doi.org/10.18653/v1/2021.acl-long.131

[18] Khanuja, S., Dandapat, S., Srinivasan, A., Sitaram, S., & Choudhury, M. (2020). GLUECoS: An evaluation benchmark for code-switched NLP. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 3575–3585). https://doi.org/10.18653/v1/2020.acl-main.329

[19] Nayak, R., & Joshi, R. (2022). L3Cube-HingCorpus and HingBERT: A code mixed Hindi-English dataset and BERT language models. In *Proceedings of the WILDRE-6 Workshop within the 13th Language Resources and Evaluation Conference* (pp. 7–12). European Language Resources Association.

[20] Joshi, A., Prabhu, A., Shrivastava, M., & Varma, V. (2016). Towards sub-word level compositions for sentiment analysis of Hindi-English code mixed text. In *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers* (pp. 2482–2491). The COLING 2016 Organizing Committee.

[21] Patwa, P., Aguilar, G., Kar, S., Pandey, S., PYKL, S., Gambäck, B., Chakraborty, T., Solorio, T., & Das, A. (2020). SemEval-2020 Task 9: Overview of sentiment analysis of code-mixed tweets. In *Proceedings of the Fourteenth Workshop on Semantic Evaluation* (pp. 774–790). International Committee for Computational Linguistics. https://doi.org/10.18653/v1/2020.semeval-1.100

[22] Srinivasan, A. (2020). MSR India at SemEval-2020 Task 9: Multilingual models can do code-mixing too. In *Proceedings of the Fourteenth Workshop on Semantic Evaluation* (pp. 951–956). International Committee for Computational Linguistics.

[23] Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., & Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 8440–8451). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.747

[24] Singh, P., & Lefever, E. (2020). Sentiment analysis for Hinglish code-mixed tweets by means of cross-lingual word embeddings. In *Proceedings of the 4th Workshop on Computational Approaches to Code Switching* (pp. 45–51). European Language Resources Association.

[25] Baroi, S. J., Singh, N., Das, R., & Singh, T. D. (2020). NITS-Hinglish-SentiMix at SemEval-2020 Task 9: Sentiment analysis for code-mixed social media text using an ensemble model. In *Proceedings of the Fourteenth Workshop on Semantic Evaluation*. International Committee for Computational Linguistics. https://doi.org/10.48550/arXiv.2007.12081

[26] Chakravarthi, B. R., Priyadharshini, R., Muralidaran, V., Jose, N., Suryawanshi, S., Sherly, E., & McCrae, J. P. (2022). DravidianCodeMix: Sentiment analysis and offensive language identification dataset for Dravidian languages in code-mixed text. *Language Resources and Evaluation, 56*(3), 765–806. https://doi.org/10.1007/s10579-022-09583-7

[27] Mehmood, K., Essam, D., Shafi, K., & Malik, M. K. (2020). An unsupervised lexical normalization for Roman Hindi and Urdu sentiment analysis. *Information Processing & Management, 57*(6), 102368. https://doi.org/10.1016/j.ipm.2020.102368

[28] Mabokela, K. R., Celik, T., & Raborife, M. (2023). Multilingual sentiment analysis for under-resourced languages: A systematic review of the landscape. *IEEE Access, 11*, 15996–16020. https://doi.org/10.1109/ACCESS.2022.3224136

[29] Plutchik, R. (1980). A general psychoevolutionary theory of emotion. In R. Plutchik & H. Kellerman (Eds.), *Emotion: Theory, research, and experience* (Vol. 1, pp. 3–33). Academic Press.

[30] Mohammad, S. M., & Turney, P. D. (2013). Crowdsourcing a word–emotion association lexicon. *Computational Intelligence, 29*(3), 436–465. https://doi.org/10.1111/j.1467-8640.2012.00460.x

[31] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?" Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 1135–1144). ACM. https://doi.org/10.1145/2939672.2939778

[32] Ribeiro, M. T., Wu, T., Guestrin, C., & Singh, S. (2020). Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 4902–4912). https://doi.org/10.18653/v1/2020.acl-main.442

[33] Zhang, W., Deng, Y., Liu, B., Pan, S. J., & Bing, L. (2024). Sentiment analysis in the era of large language models: A reality check. In *Findings of the Association for Computational Linguistics: NAACL 2024* (pp. 3881–3906). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.findings-naacl.246

[34] Zhang, R., Cahyawijaya, S., Cruz, J. C. B., Winata, G. I., & Aji, A. F. (2023). Multilingual large language models are not (yet) code-switchers. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (pp. 12567–12582). https://doi.org/10.18653/v1/2023.emnlp-main.774

[35] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830.

[36] McCallum, A., & Nigam, K. (1998). A comparison of event models for naive Bayes text classification. In *AAAI-98 Workshop on Learning for Text Categorization* (pp. 41–48). AAAI Press.

[37] Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to information retrieval*. Cambridge University Press. https://doi.org/10.1017/CBO9780511809071

[38] Kohavi, R. (1995). A study of cross-validation and bootstrap for accuracy estimation and model selection. In *Proceedings of the 14th International Joint Conference on Artificial Intelligence* (pp. 1137–1143). Morgan Kaufmann.

[39] Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement, 20*(1), 37–46. https://doi.org/10.1177/001316446002000104

[40] Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman & Hall.

[41] McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika, 12*(2), 153–157. https://doi.org/10.1007/BF02295996

[42] Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics, 33*(1), 159–174. https://doi.org/10.2307/2529310

Conflict of Interest Statement:

The authors have no conflict of interest to disclose.

Acknowledgments:

The authors thank the Department of Computer Science and Engineering, Maharana Institute of Professional Studies, Kanpur, for its support and guidance, and acknowledge the open-source communities behind Node.js, Python, scikit-learn, NumPy and Matplotlib.